

FINN 2041 CORPORATE FINANCE

2025/2026

Managing Risks

*futures,
option*



Learning objectives

- ❑ By the end of the session, you should be able to:
 - Explain reasons why firms manage risk
 - Identify and explain the different types of risk that corporate firms face
 - Identify and explain the range of tools firm can use to manage different types of risk
 - Explain why firms sometimes do not adopt risk management strategies

Required readings

- Brealey, R., Myers, S., Allen, F. and Edmans, A. (2022) [Principles of Corporate Finance](#), 14th ed., McGraw Hill: New York., Chapter 27

Why Manage Risk?

☐ Cash shortfalls

☐ Financial distress

☐ Agency costs

☐ Currency fluctuations

☐ Political instability

Insurance

- Most businesses face the possibility of a hazard that can bankrupt the company in an instant
- These risks are neither financial- or business-related and cannot be diversified
- The cost and risk of a loss due to a hazard, however, can be shared by others who share the same risk

Insurance Continued

Example

An offshore oil platform is valued at \$1 billion. Expert meteorologist reports indicate that a 1-in-10,000 chance exists that the platform may be destroyed by a storm in any year.

How can the cost of this hazard be shared?

Insurance

A large number of companies with similar risks can each contribute pay into a fund that is set aside to pay the cost should a member of this risk-sharing group experience the 1-in-10,000 loss. The other 9,999 firms may not experience a loss, but they also avoided the risk of not being compensated should a loss have occurred.

Insurance

What would the cost to each group member be for this protection?

$$\frac{1,000,000,000}{10,000} = \underline{\$100,000}$$

Insurance

- ❑ The loss of an oil platform by a storm may be 1 in 10,000. The risk, however, is larger for an insurance company since all the platforms in the same area may be insured; thus, if a storm damages one, it may damage all in the same area. The result is a much larger risk to the insurer.
- ❑ Catastrophe bonds (CAT bonds): Allow insurers to transfer their risk to bondholders by selling bonds whose cash flow payments depend on the level of insurable losses NOT occurring.

Foreign currency risks

□ Types of foreign currency risks firms face with
(especially multinational firms)

- Transaction risk 交易
 - the risk that exchange rates will move between the time that a foreign currency transaction is agreed and the point at which the transaction is completed and accounts are settled.
 - This could result in a transaction being less profitable than anticipated when expressed in the company's domestic currency.

Foreign currency risks

❑ Types of foreign currency risks firms face (especially multinational firms)

- Translation risk- 外币兑换
 - the risk that overseas assets and liabilities will appear to have changed in value merely due to the rate at which they are translated for incorporation into the domestic accounts.
 - Whilst this is not a cash movement, it will still impact of the value of the company and, therefore, on the wealth of shareholders.

Foreign currency risks

❑ Types of foreign currency risks firms face (especially multinational firms)

- Economic risk
 - the risk that the amounts received from an overseas project, expressed in the domestic currency of the company, will change due to a change in the exchange rate over time.
 - This could result in a project that was believed to be profitable becoming unprofitable even though it performed according to expectations in terms of the foreign currency.

Interest rate risks

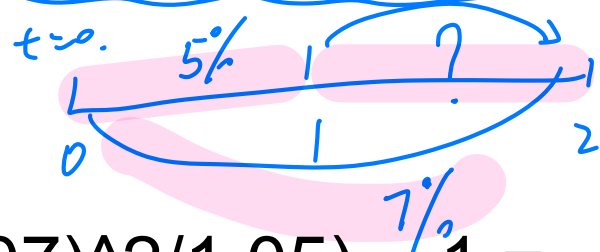
- ❑ Firms with significant floating rate debt are concerned about interest rate increases.
- ❑ Firms with a lot of fixed rate debt may lose competitive advantage if interest rates fall. ↓
- ❑ Gap exposure arises if floating rates on the same basis are revised over different periods.

Interest rate risks

□ Forward rates

- If the one-year spot interest rate is 5 percent and the two-year spot interest rate is 7 percent, what is the one-year forward interest rate one year from today.

当期



- Forward interest rate = $((1.07)^2 / 1.05) - 1 =$

9% $(1 + 7\%)^2 = (1 + \text{forward rate}) \times (1 + 5\%)$

$$\frac{1.07^2}{1.05} - 1 = \frac{(1 + 7\%)^2}{(1 + 5\%)} - 1$$

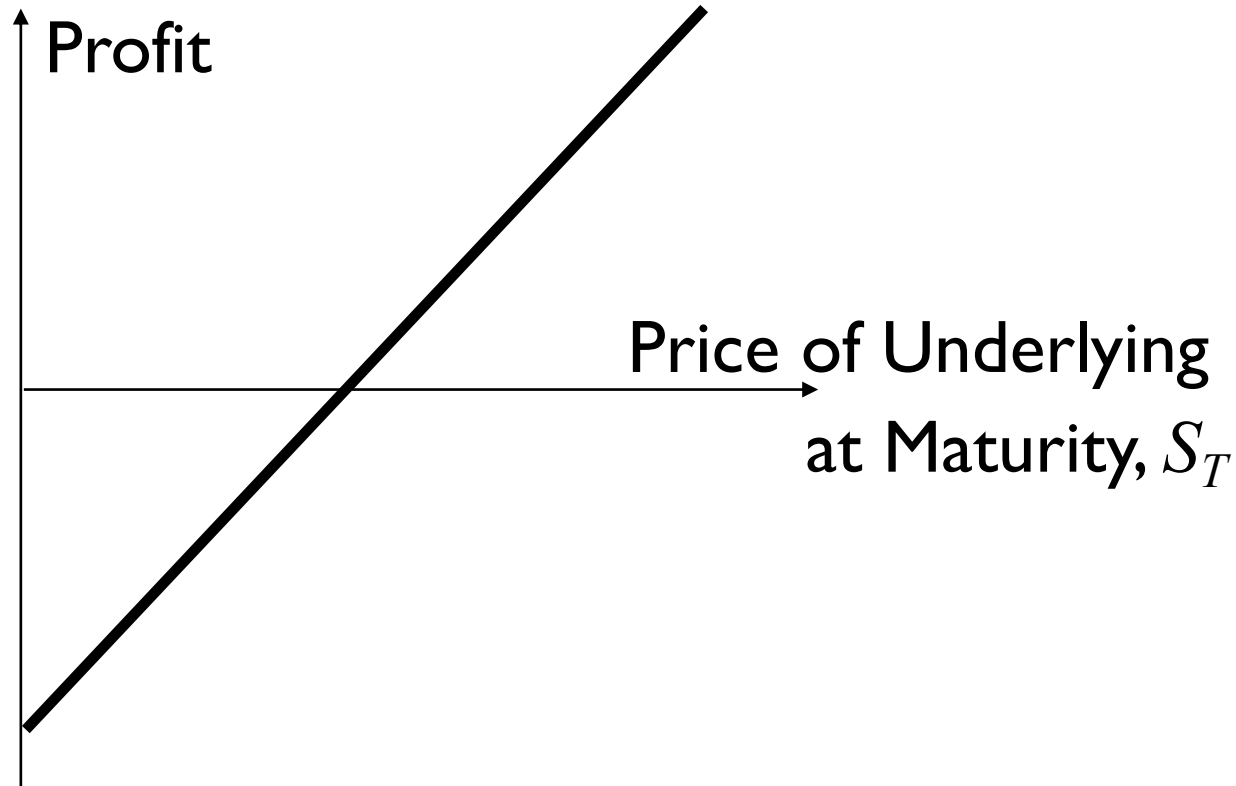
Managing risk using derivatives

□ Forward contracts

- An agreement to buy or sell a specified asset on a future date at a specified price
- No money is exchanged at initiation of contract
- Underlying asset is usually a commodity, currency or interest rate
- Usually private contracts, not designed to be traded

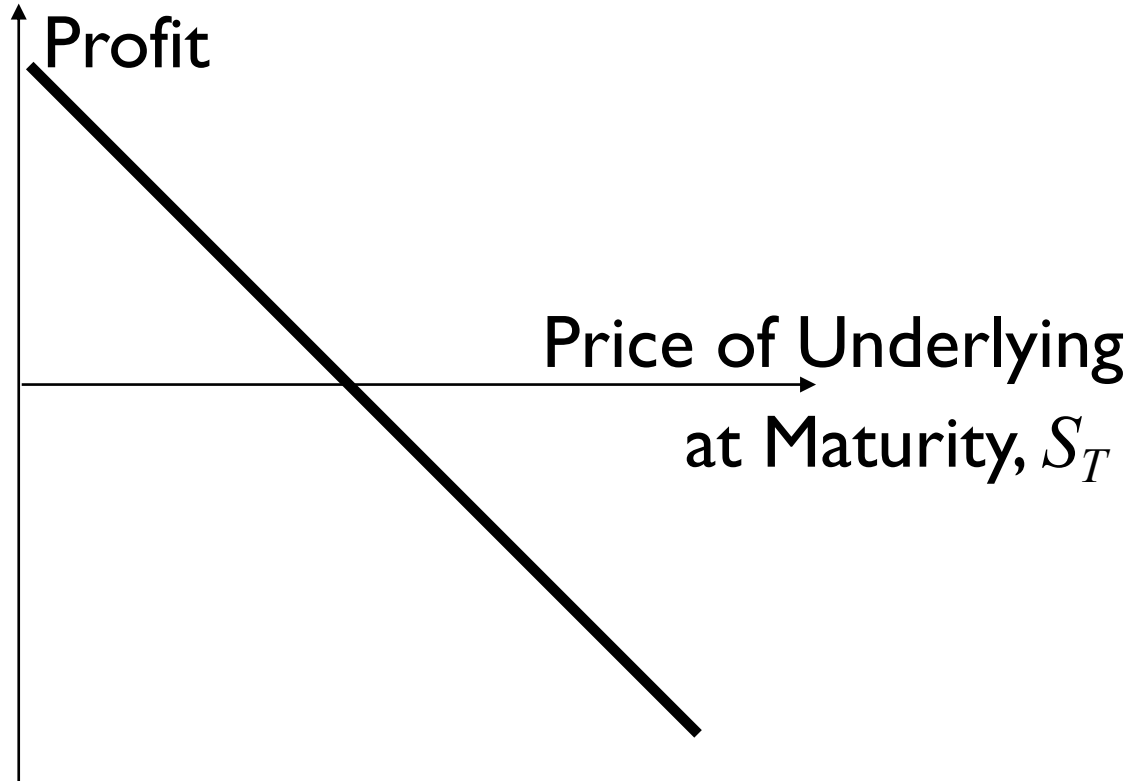
Managing risk using derivatives

Payoff from long forward position



Managing risk using derivatives

Payoff from short forward position



Managing risk using derivatives

❑ Forward contract example

- Arctic Fuels plans to deliver one million gallons of heating oil to its retail customers next January.
- Arctic worries about high heating-oil prices next winter and wants to lock in the cost of buying its supply.
- Northern Refineries is in an opposite position. It will produce heating oil next winter, but doesn't know what the oil can be sold for.

Managing risk using derivatives

❑ Forward contract example

- The two firms strike a deal: Arctic Fuels agrees to pay \$2.40 per gallon for 1 million gallons, to be paid on delivery in January.

- Arctic and Northern are now two counterparties in a forward contract.

- Regardless of what happens to future spot prices, each counterparty has locked into a price of \$2.40 per gallon.

Managing risk using derivatives

☐ Futures contracts

- Identical to a forward contract.
- But ...
- Trades are made on organized futures exchanges (e.g. Chicago Board of Trade (CBOT), Chicago Mercantile Exchange (CME), NYMEX, LIFFE (London), TFX (Tokyo), ZCE, DCE, SHFE and CFFEX (China)). All terms except price are fixed to allow trading.
- To avoid default of either counterparty
- Traders are required to post collateral (margin).
- Cash flows are exchanged on a daily basis based on the change in the price of the futures contract (marking to market).

Managing risk using derivatives

❑ Forward vs futures contracts

FORWARDS	FUTURES
Private contract between 2 parties	Exchange traded
Non-standard contract	Standard contract
Settled at end of contract	Settled daily
Delivery or final cash settlement usually occurs	Contract usually closed out prior to maturity
Some credit risk	Virtually no credit risk

Managing risk using derivatives

☒ Options



A contract that gives its owner the right (but not the obligation) to purchase or sell an asset at a fixed price at some future date

- Call Option

A financial option that gives its owner the right to buy an asset

- Put Option

A financial option that gives its owner the right to sell an asset

Managing risk using derivatives

❑ Options

- Exercising an Option

When a holder of an option enforces the agreement and buys or sells the asset at the agreed-upon price

- Strike Price (Exercise Price)

The price at which an option holder buys or sells the asset when the option is exercised

- Expiration Date

The last date on which an option holder has the right to exercise the option

Managing risk using derivatives

□ Options

- S_0 = Current stock price
- K = Strike price / exercise price
- S_T = Stock price at expiration of option
- T : Time to expiration of option
- c = Value of call option (European call option)
- p = Value of put option (European put option)

Managing risk using derivatives

□ Call option payoffs

- Long position

$$\max (S_T - K, 0)$$

- Short position

$$\min (K - S_T, 0)$$

Where S_T is the stock price at expiration, K is the exercise price, and *max (min)* is the maximum (minimum) of the two quantities in the parentheses

Managing risk using derivatives

❑ Put option payoffs

- Long position

$$\max (K - S_T, 0)$$

- Short position

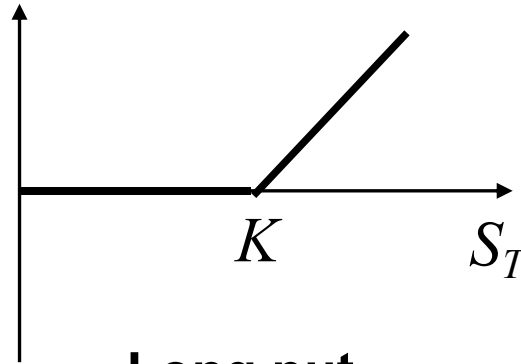
$$\min (S_T - K, 0)$$

Where S_T is the stock price at expiration, K is the exercise price, and $\max (\min)$ is the maximum (minimum) of the two quantities in the parentheses

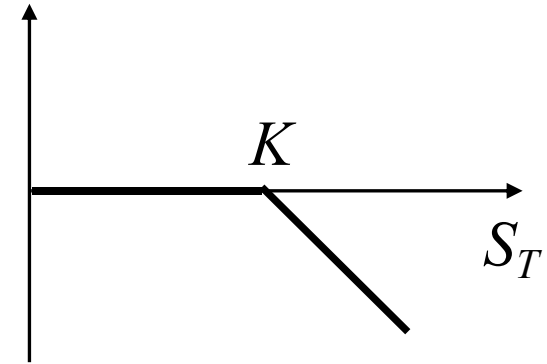
Managing risk using derivatives

Options

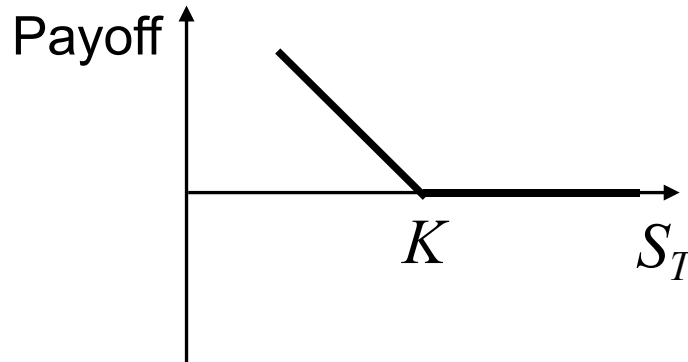
Long call



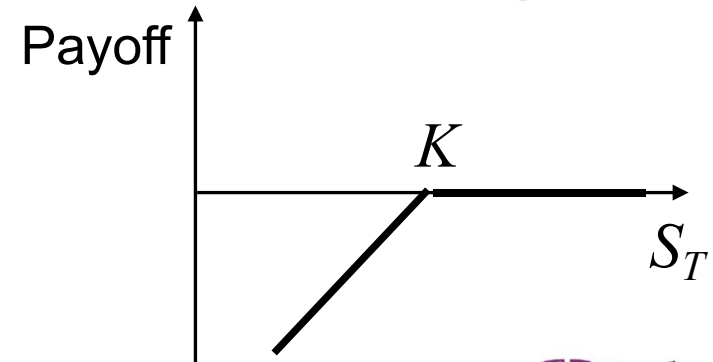
Short call



Long put



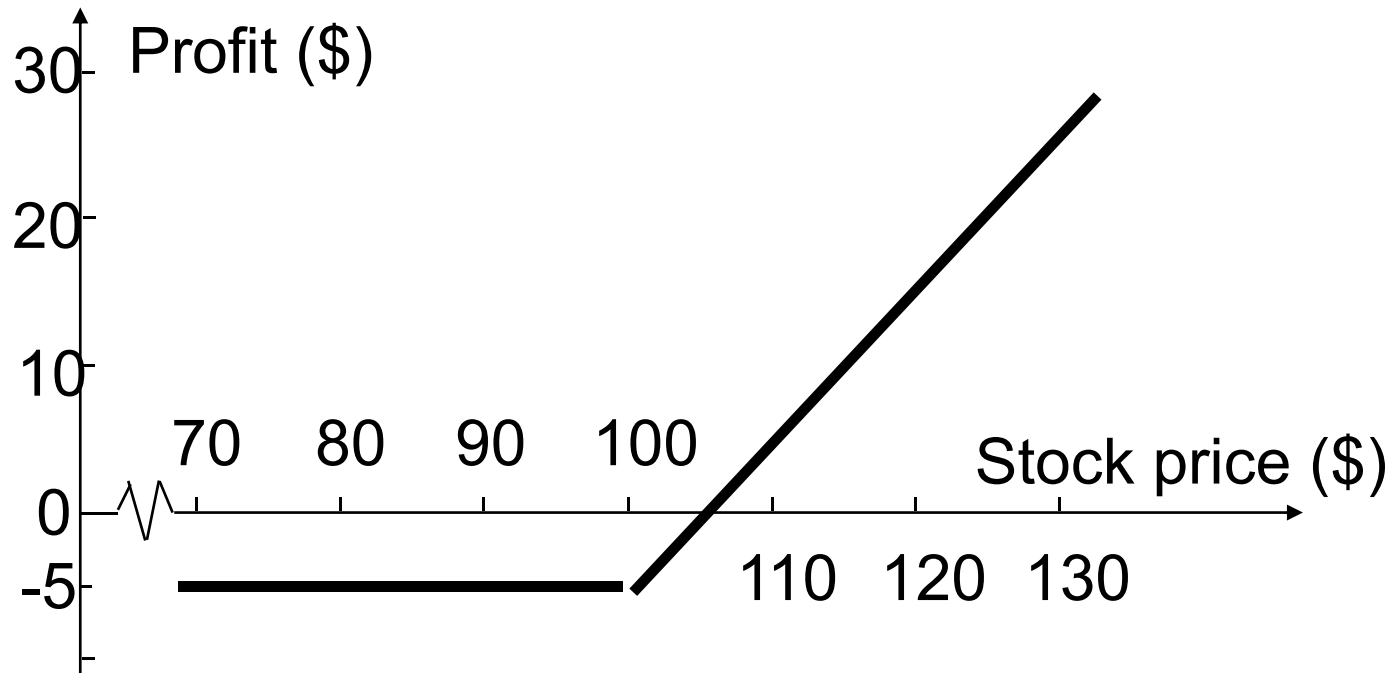
Short put



Managing risk using derivatives

□ Long Call

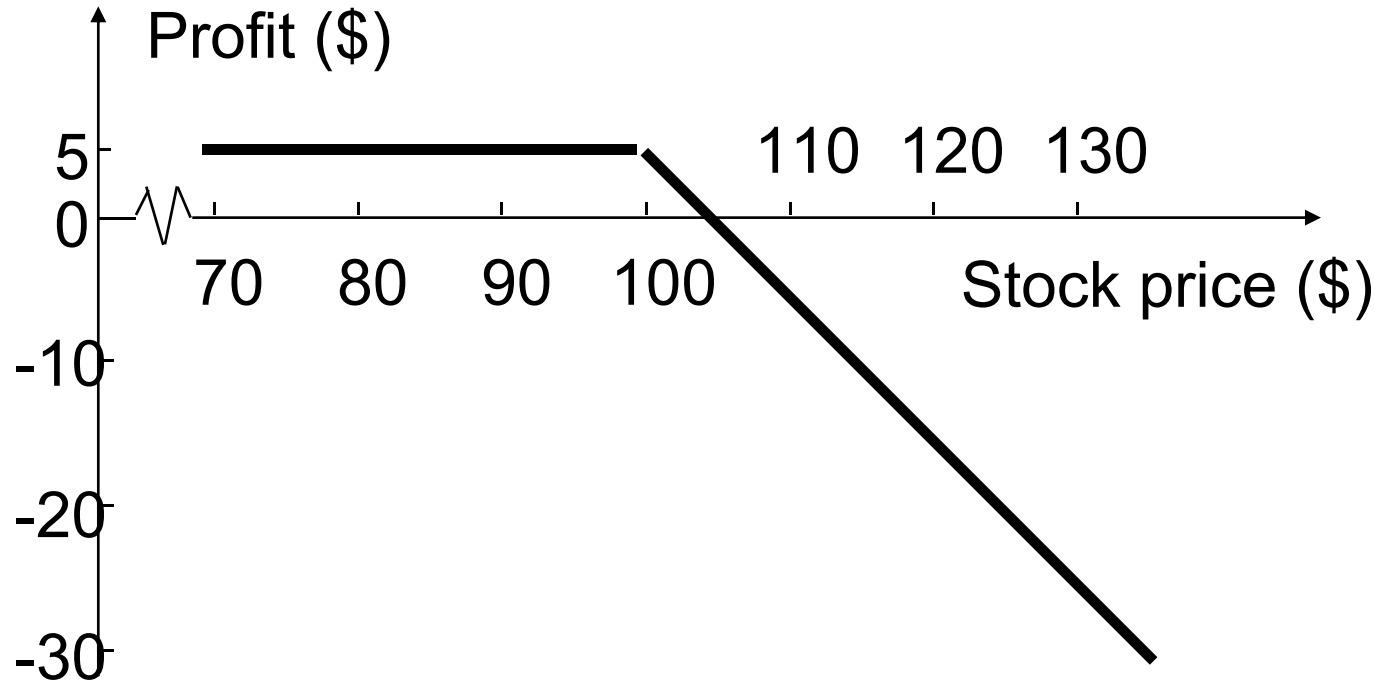
Profit from buying one European call option: option price = \$5, strike price = \$100



Managing risk using derivatives

□ Short call

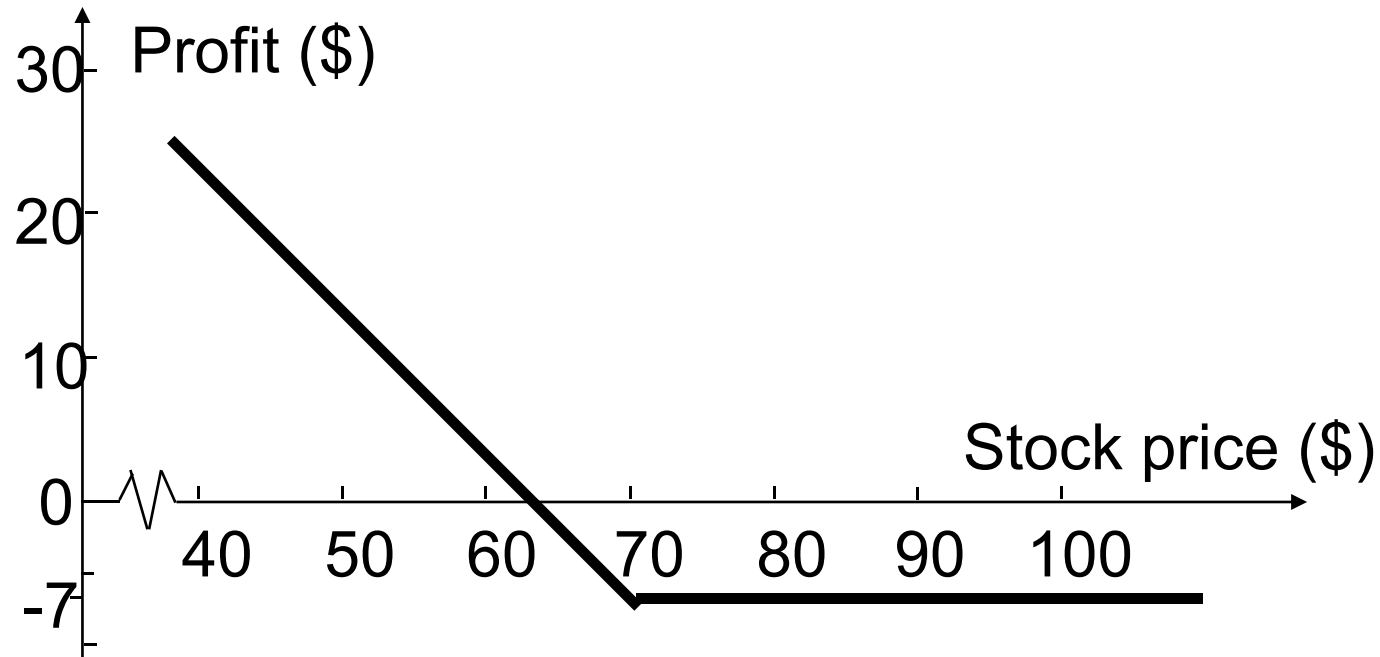
Profit from writing one European call option: option price = \$5, strike price = \$100



Managing risk using derivatives

□ Long put

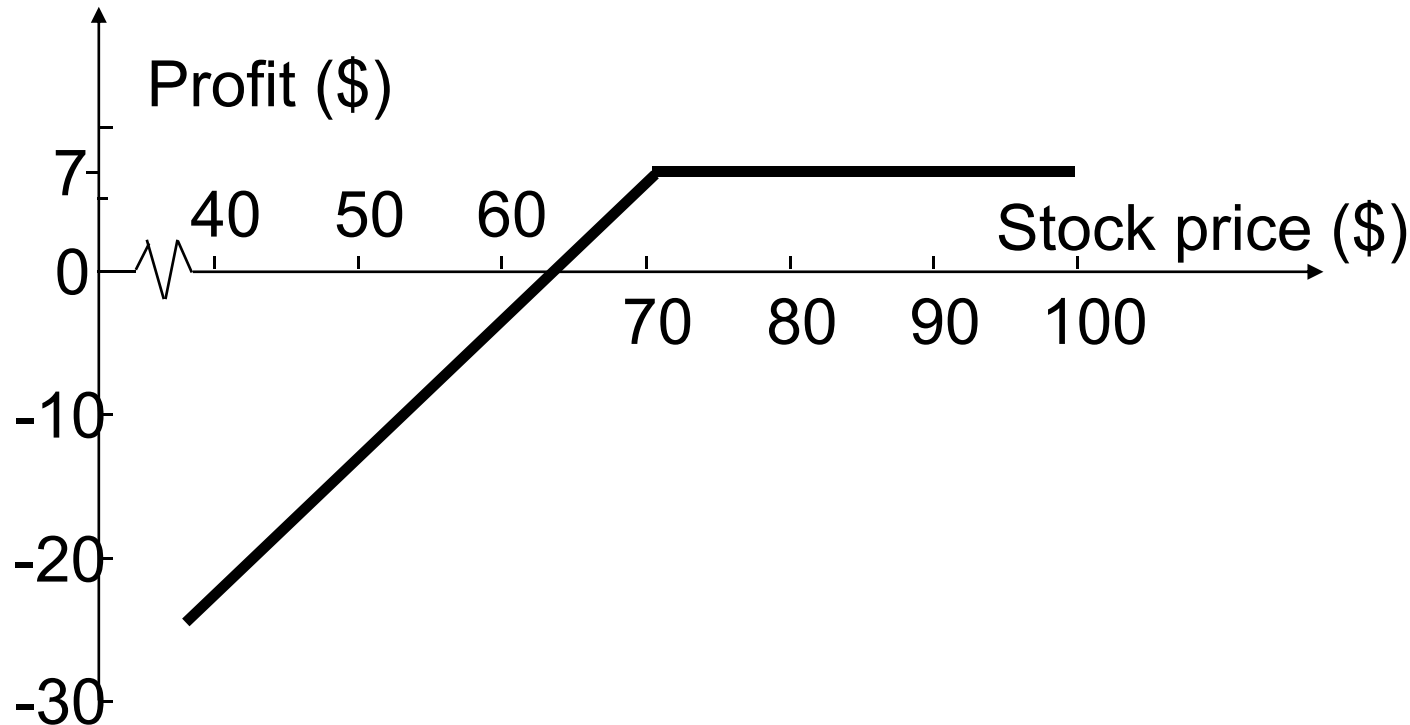
Profit from buying a European put option: option price = \$7, strike price = \$70



Managing risk using derivatives

❑ Short put

Profit from writing a European put option: option price = \$7, strike price = \$70



Managing risk using derivatives

❑ Swaps

互换

Interest Rate Swaps

- A: Swap fixed for floating payments
- B: Swap floating for fixed payments

Currency Swaps

- A: Swap obligation to pay in currency X for obligation to pay in currency Y
- B: Swap obligation to pay in currency Y for obligation to pay in currency X

Managing foreign currency risk using non-derivative contracts

☐ Dealing in home currency

- Insist all customers pay in your home currency and pay for all imports in home current
- Transfers risk to the other party
- May not be commercially acceptable

Managing foreign currency risk using non-derivative contracts

□ Leading and lagging

- Leading: If an exporter expects that the currency it is due to receive will depreciate over the next few months, it may try to obtain payment immediately. This may be achieved by offering a discount for prompt payment.

Managing foreign currency risk using non-derivative contracts

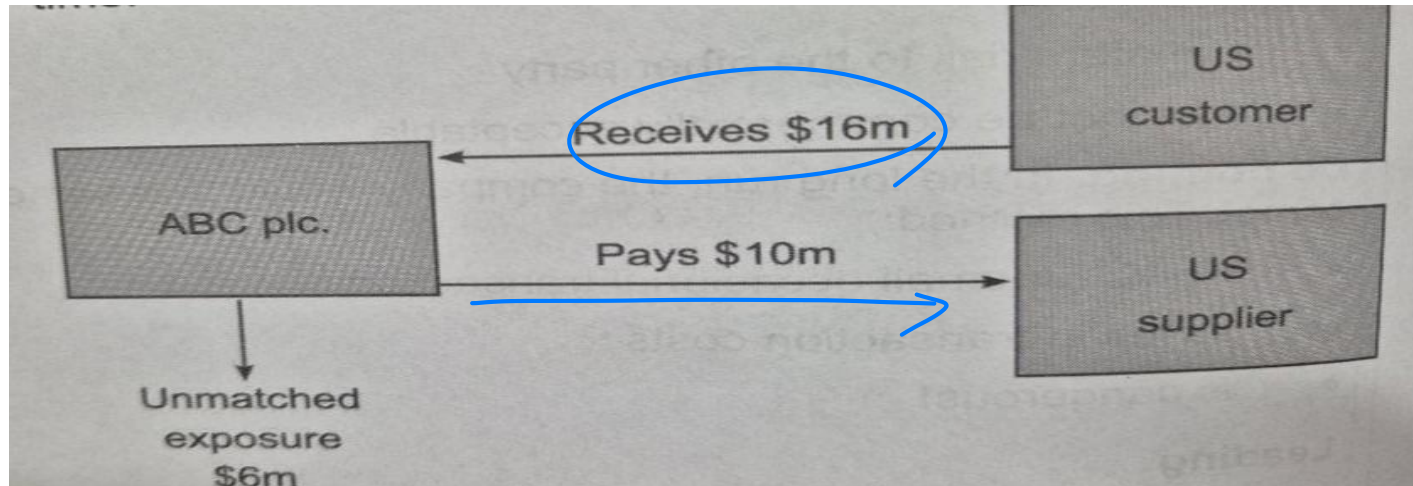
❑ Leading and lagging

- Lagging: If an importer expects the currency, it is due to pay will depreciate, it may attempt to delay payment:

Managing foreign currency risk using non-derivative contracts

❑ Matching payments and receipts

- Match receipts and payments in same currency due at the same time against each other.
- Then deal with the foreign exchange market for the unmatched total transaction



Managing foreign currency risk using non-derivative contracts

- Maintaining foreign currency bank accounts
 - Operates as a permanent matching process
 - Exposure to exchange risk is limited to the net balance on the account

Managing foreign currency risk using non-derivative contracts

❑ Matching assets and liabilities

- For example: Borrowing in a foreign currency and using revenue receipts later to pay the loan

Managing risks associated with working capital

□ Accounts payables

- Negotiating with suppliers for discounts on early payments in order to reduce overall cost and optimize cash flow
- Optimising payment terms by negotiating favourable terms such as extending due dates without penalties. This helps to maintain liquidity and align with cash flow cycles
- Undertaking regular monitoring and reconciliation to avoid duplicates and missed invoices
- Automating account payable processes to reduce human error and speed up approvals

Managing risks associated with working capital

□ Accounts receivables

- Offer discounts for early payment of invoices Although the company would be reducing the sales it would be saving on overdraft interest.
- Invoice discounting – By selling the trade receivables – particularly late payers to an invoice discounting company Cable receive its cash before the customers pay, however this does cost a fee.

Why a company might choose not to hedge foreign currency risks

- ❑ Cost implications
- ❑ The exposure may also not be material
- ❑ Complexities with using and accounting for derivatives
- ❑ Attitude to risk may lead the firm to decide to leave the upside potential open
- ❑ Shareholders may already be fully diversified, so there may be no benefit for them from hedging

Political risks

□ This can be defined as:

‘the possibility of a multinational company being significantly affected by political events in a host country or a change in the political relationships between a host country and one or more other countries’.

□ It can be assessed at a macro or micro level through a range of methods.

Political risks

□ Assessing political risks

- Checklist approach with the aim of producing a political risk index
- Quantitative analysis techniques, such as sensitivity analysis or discriminant analysis
- Inspection visits or 'grand tours' to countries being considered for FDI

Political risks

- ❑ Political risk management policies:
 - Insurance against political risk
 - Negotiation of concession agreements with host governments
 - Structuring of operating policies, including locating different stages of construction in different countries